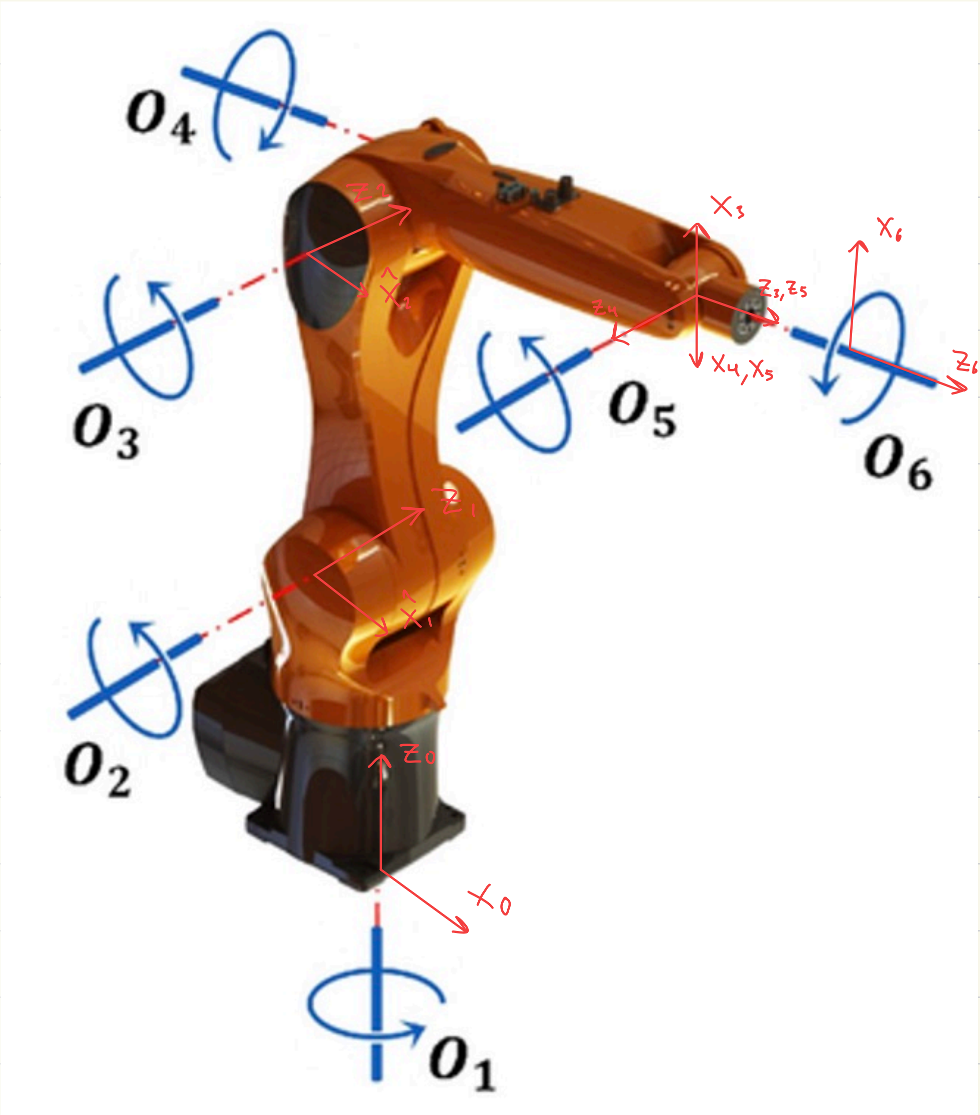
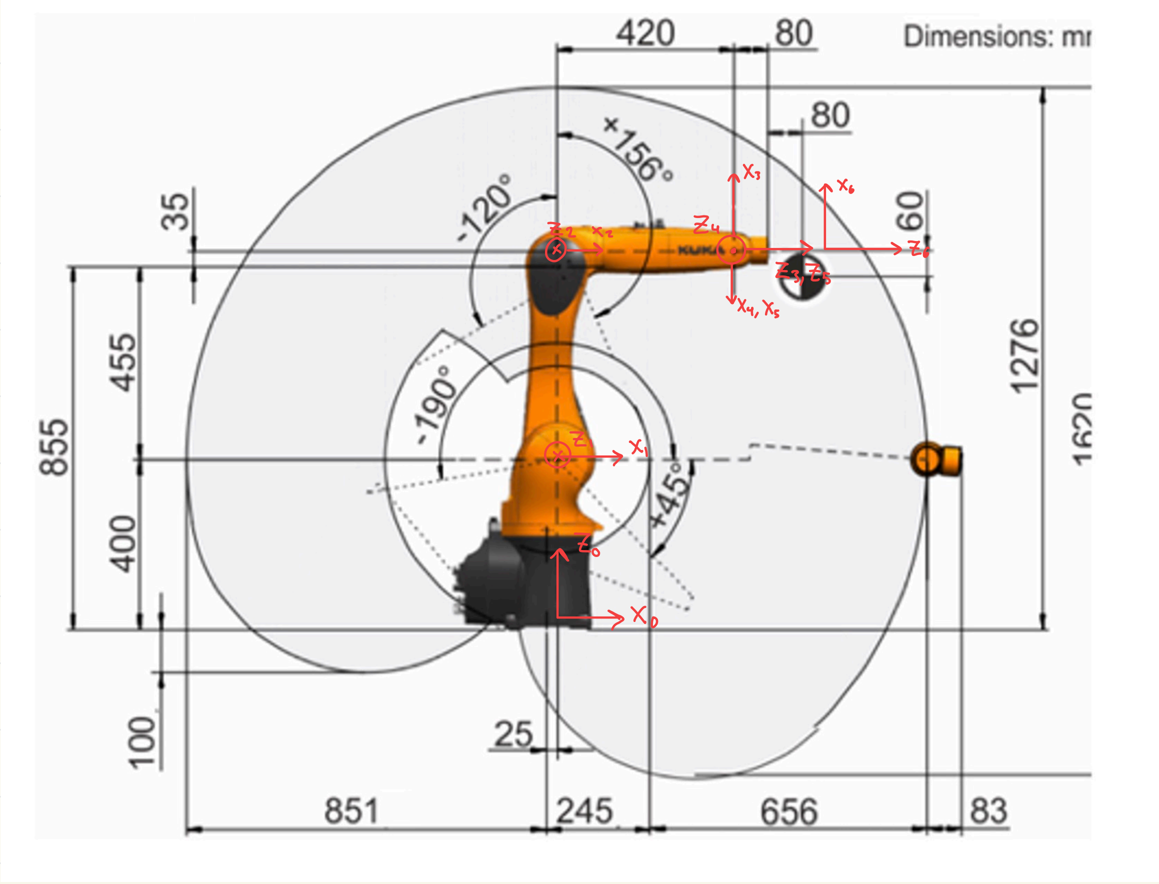
Problem 2





Dimensions: mr
420
80
80
35
60
855
455
400
100
1276
1620
-120°
+156°
-190°
+45°
25
851
245
656
83
KUKA
z_2 x_2
x_3
z_4
x_4, x_5
z_3, z_5
x_6
z_6
z_1 x_1
z_0
x_0

DH table

$link_i$	a_i	α_i	d_i	θ_i
1	0.025	$-\frac{\pi}{2}$	0.4	θ_1^*
2	0.455	0	0	θ_2^*
3	0.035	$-\frac{\pi}{2}$	0	θ_3^*
4	0	$\frac{\pi}{2}$	0.42	θ_4^*
5	0	$-\frac{\pi}{2}$	0	θ_5^*
6	0	0	0.08	θ_6^*

P3

$$T_i^{i-1} = \begin{bmatrix} C\theta_i & -S\theta_i C\alpha_i & S\theta_i S\alpha_i & a_i C\theta_i \\ S\theta_i & C\theta_i C\alpha_i & -C\theta_i S\alpha_i & a_i S\theta_i \\ 0 & S\alpha & C\alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Row 1: $a_1 = 0.025,\ \alpha_1 = -\frac{\pi}{2},\ d_1 = 0.4,\ \theta_1 = \theta_1^*$

$$T_1^0 = \begin{bmatrix} C\theta_1 & 0 & -S\theta_1 & 0.025C\theta_1 \\ S\theta_1 & 0 & C\theta_1 & 0.025S\theta_1 \\ 0 & -1 & 0 & 0.4 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Row 2: $a_2 = 0.455,\ \alpha_2 = 0,\ d_2 = 0,\ \theta_2 = \theta_2^*$

$$T_2^1 = \begin{bmatrix} C\theta_2 & -S\theta_2 & 0 & 0.455C\theta_2 \\ S\theta_2 & C\theta_2 & 0 & 0.455S\theta_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Row 3: $a_3 = 0.035,\ \alpha_3 = -\frac{\pi}{2},\ d_3 = 0,\ \theta_3 = \theta_3^*$

$$T_3^2 = \begin{bmatrix} C\theta_3 & 0 & -S\theta_3 & 0.035C\theta_3 \\ S\theta_3 & 0 & C\theta_3 & 0.035S\theta_3 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Row 4: $a_4 = 0,\ \alpha_4 = \frac{\pi}{2},\ d_4 = 0.42,\ \theta_4 = \theta_4^*$

$$T_4^3 = \begin{bmatrix} C\theta_4 & 0 & S\theta_4 & 0 \\ S\theta_4 & 0 & -C\theta_4 & 0 \\ 0 & 1 & 0 & 0.42 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Row 5: $a_5 = 0,\ \alpha_5 = -\frac{\pi}{2},\ d_5 = 0,\ \theta_5 = \theta_5^*$

$$T_5^4 = \begin{bmatrix} C\theta_5 & 0 & -S\theta_5 & 0 \\ S\theta_5 & 0 & C\theta_5 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Row 6: $a_6 = 0$, $\alpha_6 = 0$, $d_6 = 0.08$, $\theta_6 = \theta_6^*$

$$T_6^5 = \begin{bmatrix} C\theta_6 & -S\theta_6 & 0 & 0 \\ S\theta_6 & C\theta_6 & 0 & 0 \\ 0 & 0 & 1 & 0.08 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\vec{q} = [\theta_1, \theta_2, \theta_3, \theta_4, \theta_5, \theta_6]^T$$

Multiplying to get T_6^0

$$T_6^0 = T_1^0 \cdot T_2^1 \cdot T_3^2 \cdot T_4^3 \cdot T_5^4 \cdot T_6^5$$

through matrix multiplication calculator:

$$T_6^0 = \begin{bmatrix}
C_6(-C_1S_{23}S_5 + C_5(C_1C_{23}C_4 + S_1S_4)) + S_6(-C_1C_{23}S_4 + C_4S_1) & C_6(-C_1C_{23}S_4 + C_4S_1) - S_6(-C_1S_{23}S_5 + C_5(C_1C_{23}C_4 + S_1S_4)) & -C_1C_5S_{23} - S_5(C_1C_{23}C_4 + S_1S_4) & 0.455C_1C_2 + 0.035C_1C_{23} - 0.42C_1S_{23} - 0.08C_1C_5S_{23} - 0.08C_1C_{23}C_4S_5 - 0.08S_1S_4S_5 + 0.025C_1 \\
C_6(C_5(-C_1S_4 + C_{23}C_4S_1) - S_1S_{23}S_5) - S_6(C_1C_4 + C_{23}S_1S_4) & -C_6(C_1C_4 + C_{23}S_1S_4) - S_6(C_5(-C_1S_4 + C_{23}C_4S_1) - S_1S_{23}S_5) & -C_5S_1S_{23} - S_5(-C_1S_4 + C_{23}C_4S_1) & 0.455C_2S_1 + 0.035C_{23}S_1 - 0.42S_1S_{23} - 0.08C_5S_1S_{23} - 0.08C_{23}C_4S_1S_5 + 0.08C_1S_4S_5 + 0.025S_1 \\
-C_6(C_{23}S_5 + C_4C_5S_{23}) + S_{23}S_4S_6 & C_6S_{23}S_4 + S_6(C_{23}S_5 + C_4C_5S_{23}) & -C_{23}C_5 + C_4S_{23}S_5 & -0.455S_2 - 0.035S_{23} - 0.42C_{23} - 0.08C_{23}C_5 + 0.08C_4S_{23}S_5 + 0.4 \\
0 & 0 & 0 & 1
\end{bmatrix}$$

Intermediate Poses for $q_{ready} = \left[0, -\frac{\pi}{3}, \frac{\pi}{3}, 0, \frac{\pi}{4}, 0\right]^T$

$$T_1^0 = \begin{bmatrix} 1 & 0 & 0 & 0.025 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0.4 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_2^0 = \begin{bmatrix} 1 & 0 & 0 & 0.025 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0.4 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0.5 & 0.866 & 0 & 0.2275 \\ -0.866 & 0.5 & 0 & -0.394 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0.5 & 0.866 & 0 & 0.2525 \\ 0 & 0 & 1 & 0 \\ 0.866 & -0.5 & 0 & 0.794 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_3^0 = \begin{bmatrix} 0.5 & 0.866 & 0 & 0.2525 \\ 0 & 0 & 1 & 0 \\ 0.866 & -0.5 & 0 & 0.794 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0.866 & 0 & 0.5 & 0.0303 \\ -0.5 & 0 & 0.866 & -0.0175 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 & 0.2525 \\ 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0.829 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_4^0 = \begin{bmatrix} 0 & 0 & 1 & 0.2525 \\ 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0.829 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0.42 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0.6725 \\ 0 & 0 & -1 & 0 \\ -1 & 0 & 0 & 0.829 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_5^0 = \begin{bmatrix} 0 & 1 & 0 & 0.6725 \\ 0 & 0 & -1 & 0 \\ -1 & 0 & 0 & 0.829 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0.7071 & 0 & -0.7071 & 0 \\ 0.7071 & 0 & 0.7071 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0.7071 & 0 & 0.7071 & 0.6725 \\ 0 & 1 & 0 & 0 \\ -0.7071 & 0 & 0.7071 & 0.829 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_6^0 = \begin{bmatrix} 0.7071 & 0 & 0.7071 & 0.6725 \\ 0 & 1 & 0 & 0 \\ -0.7071 & 0 & 0.7071 & 0.829 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0.08 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -0.7071 & 0 & 0.7071 & 0.7291 \\ 0 & -1 & 0 & 0 \\ 0.7071 & 0 & 0.7071 & 0.8856 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

P4

$$q(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3 + a_4 t^4 + a_5 t^5$$
$$\dot{q}(t) = a_1 + 2a_2 t + 3a_3 t^2 + 4a_4 t^3 + 5a_5 t^4$$
$$\ddot{q}(t) = 2a_2 + 6a_3 t + 12a_4 t^2 + 20a_5 t^3$$

T: target angle

Starting conditions:

$q(0) = q_0$, $\dot{q}(0) = 0$, $\ddot{q}(0) = 0$ ← Starting at rest

$q(T) = q_f$, $\dot{q}(T) = 0$, $\ddot{q}(T) = 0$ ← ends at rest

@ $t = 0$

$$q(0) = a_0 = q_0$$
$$\dot{q}(0) = a_1 = 0$$
$$\ddot{q}(0) = 2a_2 = 0 \Rightarrow a_2 = 0$$

@ $t = T$

$$q(T) = 0 + 0 + 0 + a_3 T^3 + a_4 T^4 + a_5 T^5 = q_f - q_0 = \Delta q$$
$$\dot{q}(T) = 3a_3 T^2 + 4a_4 T^3 + 5a_5 T^4 = 0$$
$$\ddot{q}(T) = 6a_3 T + 12a_4 T^2 + 20a_5 T^3 = 0$$

3 X 3 matrix system:

$$\begin{bmatrix} T^3 & T^4 & T^5 \\ 3T^2 & 4T^3 & 5T^4 \\ 6T & 12T^2 & 20T^3 \end{bmatrix} \begin{bmatrix} a_3 \\ a_4 \\ a_5 \end{bmatrix} = \begin{bmatrix} \Delta q \\ 0 \\ 0 \end{bmatrix}$$

Scientific calculator's 3x3 matrix solver:

$$\boxed{\begin{array}{l} a_3 = \dfrac{10\Delta q}{T^3}, \quad a_4 = -\dfrac{15\Delta q}{T^4}, \quad a_5 = \dfrac{6\Delta q}{T^5} \\ a_2 = 0, \; a_1 = 0, \; a_0 = q_0 \end{array}}$$